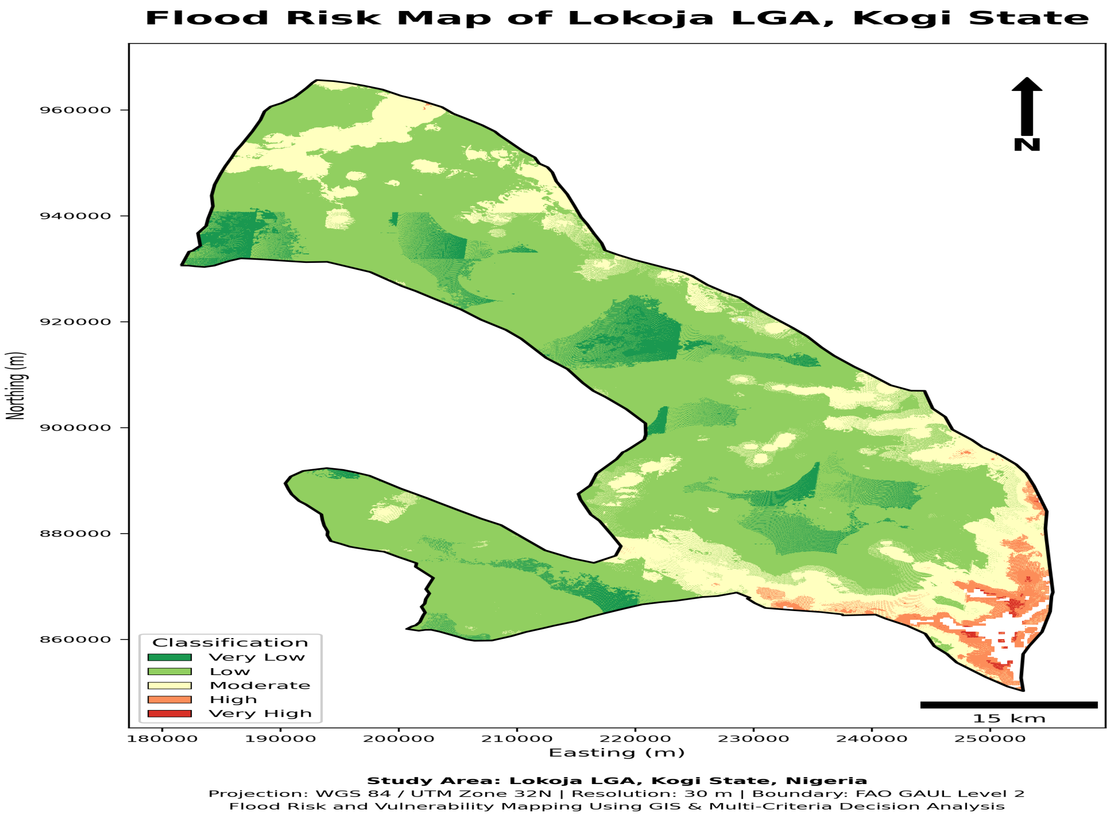
Flood Risk and Vulnerability Mapping Using GIS & Multi-Criteria Decision Analysis

Lokoja LGA, Kogi State, Nigeria



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Geospatial Planning Portfolio Project

1. Project Overview

Lokoja, the capital of Kogi State, occupies a strategically important environment influenced by major river systems and low-lying terrain. The project evaluates spatial flood hazard, human vulnerability and infrastructure exposure across Lokoja Local Government Area using Geographic Information Systems, remote sensing and Multi-Criteria Decision Analysis.

The analysis was designed to answer three planning questions: where are the most flood-prone locations, where does human settlement convert physical hazard into significant risk, and which critical infrastructure assets are exposed to the highest hazard categories?

Indicator	Result
Study area	3,406.82 km²
Estimated population (2020)	253,854
High + Very High hazard	1,305.10 km² (38.97%)
High + Very High final risk	110.39 km² (3.30%)
Population in High + Very High risk	71,195 (28.05%)
AHP Consistency Ratio	0.0093

2. Data and Methods

Spatial datasets: FAO GAUL Level-2 administrative boundary; Copernicus DEM; Sentinel-2 imagery; CHIRPS rainfall; WorldPop 2020 population; SoilGrids topsoil clay; MERIT Hydro; Dynamic World land-use/land-cover; Google Open Buildings; and OpenStreetMap road, educational and health infrastructure.

All analytical raster factors were standardized to a common 30 m grid in WGS 84 / UTM Zone 32N. Seven continuous hazard factors were reclassified using Lokoja-specific distribution thresholds, while LULC was reclassified using hydrologically meaningful susceptibility scores. AHP pairwise comparison was used to derive criterion weights. The consistency ratio was 0.0093, indicating acceptable internal consistency.

Flood-conditioning factor	AHP Weight (%)
Distance to Drainage	24.58
Elevation	20.29
Rainfall	15.85
Slope	12.18
Drainage Density	9.79
LULC	7.37
Clay Content	6.23
NDVI	3.71

Hazard model: standardized flood-conditioning scores were combined using weighted linear combination. **Vulnerability model:** native WorldPop population counts were converted to population density (people/km^2), log-transformed and normalized. **Final risk:** the continuous hazard index was combined with the population vulnerability surface. Infrastructure exposure was assessed separately by intersecting/sampling road, education and health assets against flood-hazard classes.

3. Flood Hazard Assessment

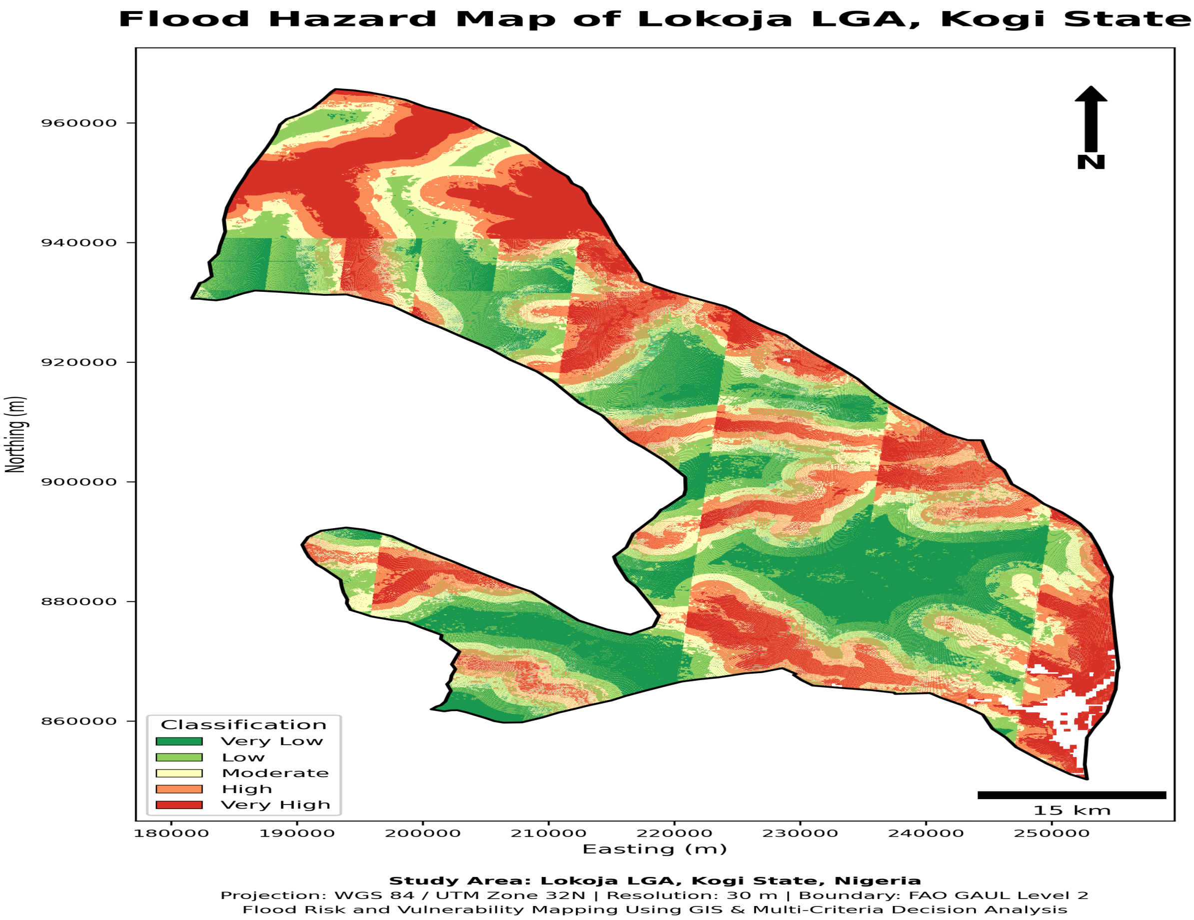
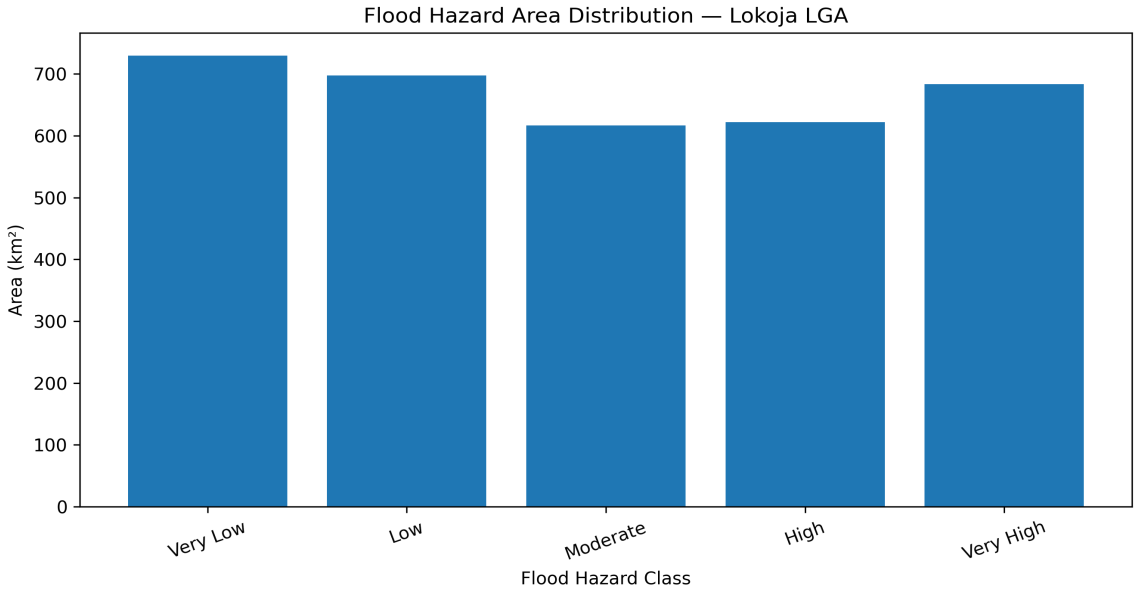


Figure 1. Final five-class flood-hazard map of Lokoja LGA.

The hazard model identified approximately 1,305.10 km², equivalent to 38.97% of the classified study area, as High or Very High flood hazard. The spatial pattern reflects the combined influence of drainage proximity, low elevation, rainfall, slope and other conditioning factors.



4. Population Vulnerability and Final Flood Risk

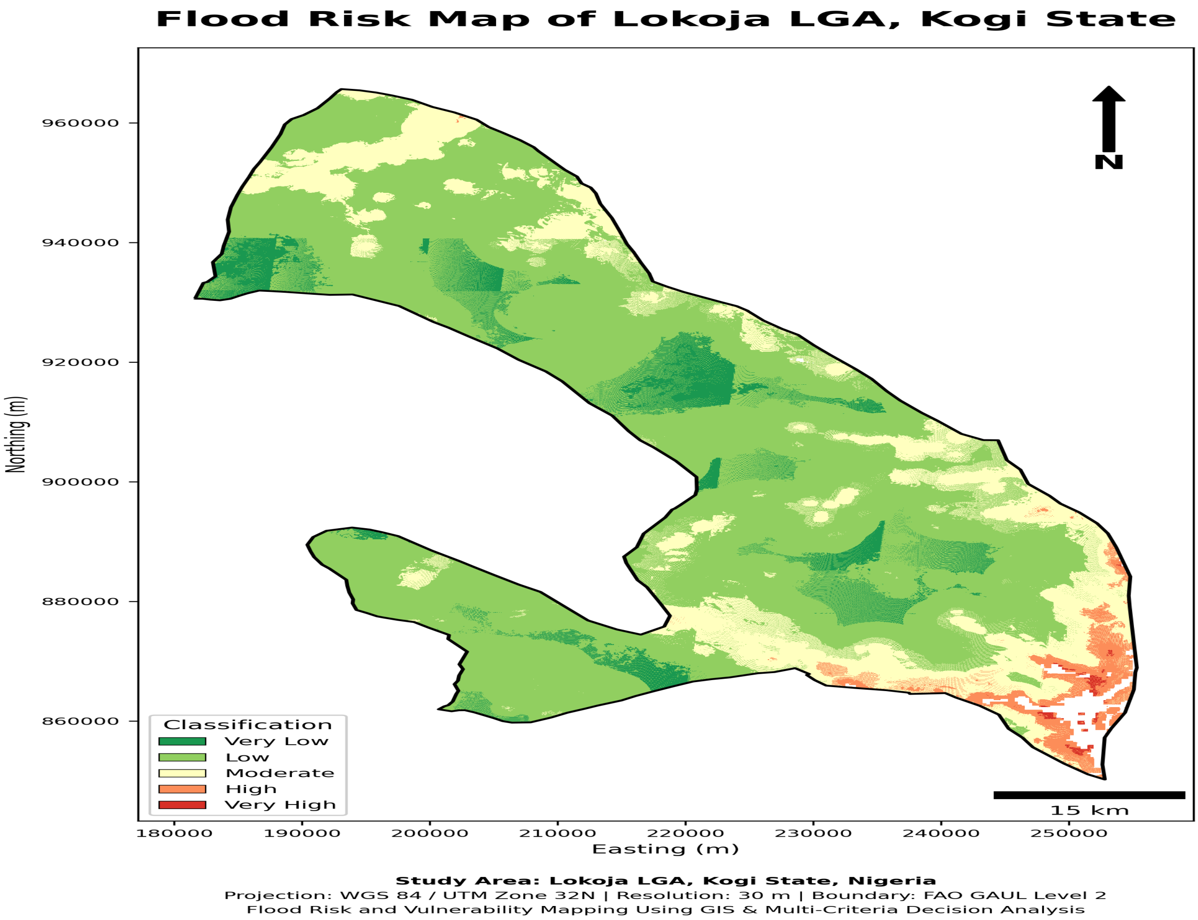
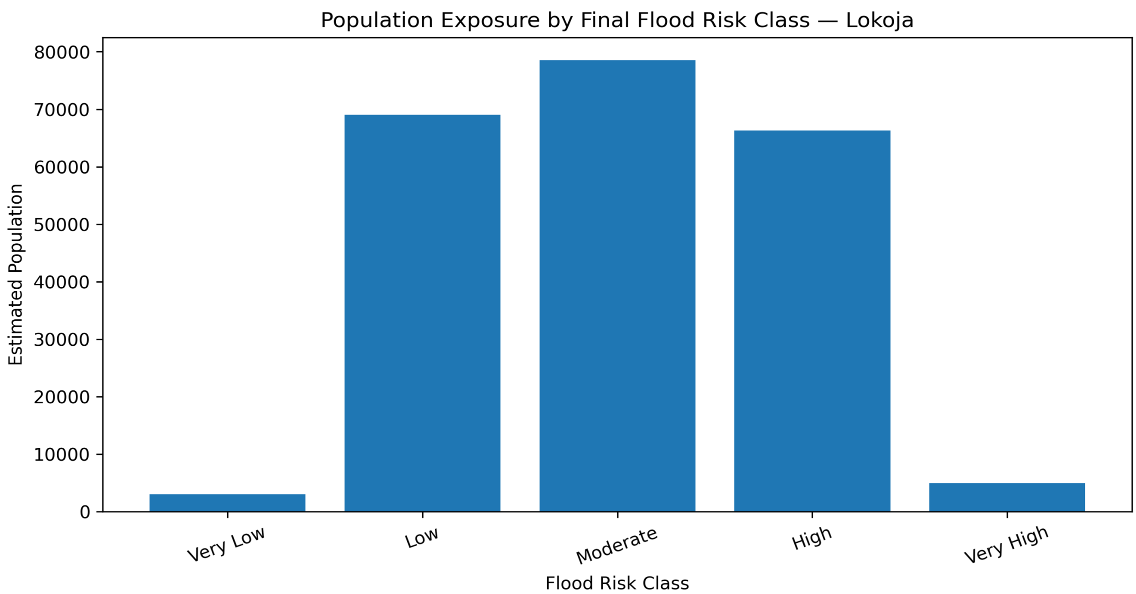


Figure 2. Final population-informed flood-risk classification.

The final risk model classified 110.39 km² (3.30% of the classified risk area) as High or Very High risk. Although this land footprint is comparatively small, it contains a disproportionately large concentration of people: approximately 71,195 residents, equivalent to 28.05% of the estimated 2020 population.



5. Infrastructure Exposure

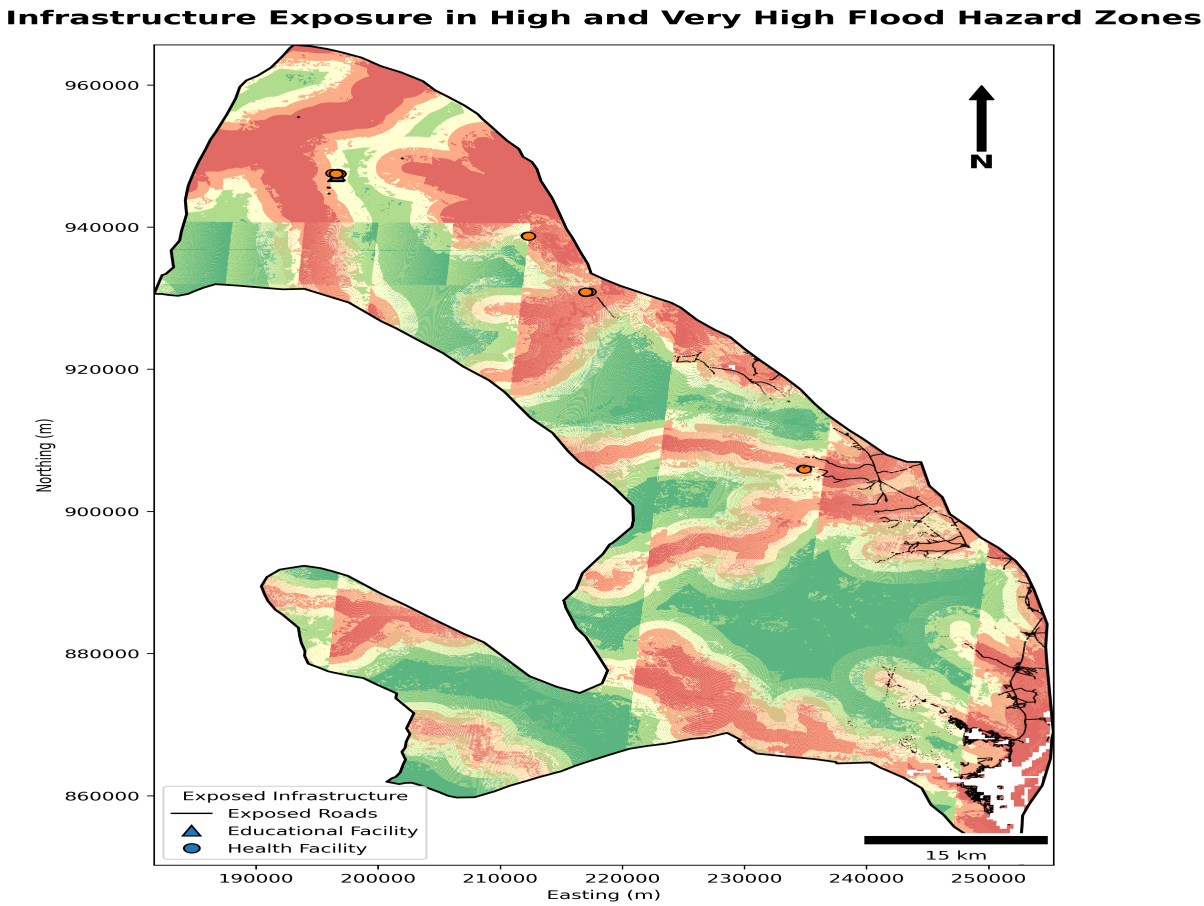
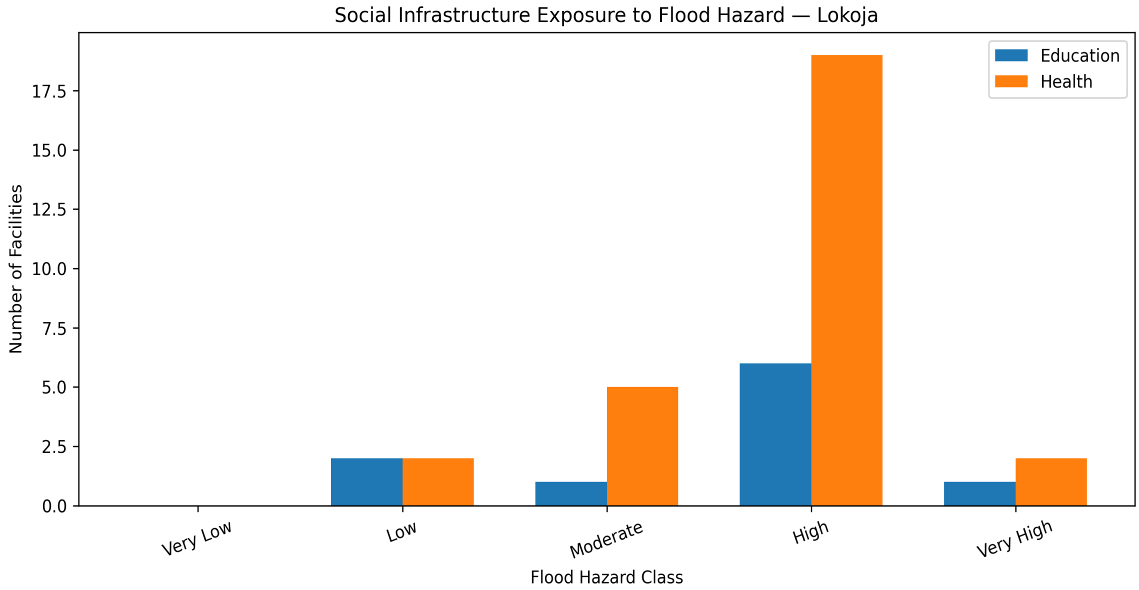


Figure 3. Road and social-infrastructure assets located within High and Very High flood-hazard zones.

The exposure analysis identified approximately 773.93 km of classified road network, 7 educational facilities and 21 health facilities within High or Very High flood-hazard zones. These results highlight potential disruption to mobility, emergency access, education and health-care delivery during major flood events.



6. Planning Implications and Recommendations

• **Risk-sensitive development control:** integrate High and Very High hazard/risk zones into development-permit reviews, land-use zoning and setback enforcement.

• **Drainage and floodplain management:** prioritize drainage maintenance, river-channel monitoring and protection of natural flood-storage areas in identified hotspots.

• **Critical infrastructure resilience:** assess the exposed road sections, schools and health facilities for elevation, access redundancy, drainage improvement and emergency planning.

• **Targeted preparedness:** use the population-risk hotspots to prioritize early-warning communication, evacuation planning and emergency shelters.

• **Continuous geospatial monitoring:** update the analysis periodically as new population, land-cover, infrastructure and hydrological data become available.

7. Key Project Outputs

The project produced permanent Earth Engine factor stacks, standardized MCDA criteria, AHP weights and consistency metrics, continuous Flood Hazard Index, final flood-hazard classes, population vulnerability index, final flood-risk index and classes, population exposure statistics, infrastructure exposure GeoPackage and CSV tables, professional maps, charts, infographic and this technical report.

8. Software and Technical Environment

Google Earth Engine was used for satellite, environmental and raster modelling; Google Colab and Python were used for OSM infrastructure acquisition, spatial exposure analysis, statistics, visualization and report generation. Core Python libraries included GeoPandas, Rasterio, OSMnx, Pandas and Matplotlib.

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